

BUSFIN 8200: Problem Set 1

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1 Question 1

1.1 Part (a)

Assume $D_t > 0$. By definition, the gross return on equity is

$$R_{e,t+1} = \frac{P_{t+1} + D_{t+1}}{P_t}. \quad (1)$$

Taking logs,

$$\begin{aligned} \log(R_{e,t+1}) &= \log\left(\frac{P_{t+1} + D_{t+1}}{P_t}\right) \\ &= \log\left(\frac{P_{t+1} + D_{t+1}}{P_{t+1}} \cdot P_{t+1}\right) - \log(P_t) \\ &= \log\left[\left(1 + \frac{D_{t+1}}{P_{t+1}}\right) P_{t+1}\right] - \log(P_t) \\ &= \log(P_{t+1}) + \log\left(1 + \frac{D_{t+1}}{P_{t+1}}\right) - \log(P_t). \end{aligned}$$

Define $r_{e,t+1} \equiv \log(R_{e,t+1})$, $p_t \equiv \log(P_t)$, and $d_t \equiv \log(D_t)$. Since $\frac{D_{t+1}}{P_{t+1}} = e^{\log(D_{t+1}/P_{t+1})} = e^{d_{t+1}-p_{t+1}}$, we can write

$$r_{e,t+1} = p_{t+1} + \log\left(1 + e^{d_{t+1}-p_{t+1}}\right) - p_t.$$

Define $f(x) \equiv \log(1 + e^x)$, where $x = d_{t+1} - p_{t+1}$. We approximate this function around the average log dividend-price ratio $\bar{d} - \bar{p}$ using a first-order Taylor approximation,

$$f(x) \approx f(\bar{d} - \bar{p}) + f'(\bar{d} - \bar{p})(x - \bar{d} - \bar{p}),$$

so that

$$f(d_{t+1} - p_{t+1}) \approx f(\bar{d} - \bar{p}) + f'(\bar{d} - \bar{p})[(d_{t+1} - p_{t+1}) - \bar{d} - \bar{p}].$$

Since $f(x) = \log(1 + e^x)$, we have $f'(x) = \frac{e^x}{1 + e^x}$, so

$$f'(\bar{d} - \bar{p}) = \frac{e^{\bar{d} - \bar{p}}}{1 + e^{\bar{d} - \bar{p}}}.$$

Define $\kappa \equiv \frac{1}{1 + e^{\bar{d} - \bar{p}}}$. Then $1 - \kappa = \frac{e^{\bar{d} - \bar{p}}}{1 + e^{\bar{d} - \bar{p}}}$, so $f'(\bar{d} - \bar{p}) = 1 - \kappa$.

Hence,

$$\begin{aligned} f(d_{t+1} - p_{t+1}) &\approx f(\overline{d-p}) + (1-\kappa) \left[(d_{t+1} - p_{t+1}) - \overline{d-p} \right] \\ &= f(\overline{d-p}) - (1-\kappa)\overline{d-p} + (1-\kappa)(d_{t+1} - p_{t+1}). \end{aligned}$$

Define $\kappa_0 \equiv f(\overline{d-p}) - (1-\kappa)\overline{d-p}$, a constant. Since $f(\overline{d-p}) = \log(1+e^{\overline{d-p}})$ and $\kappa = \frac{1}{1+e^{\overline{d-p}}}$, we have $1+e^{\overline{d-p}} = \frac{1}{\kappa}$, so

$$f(\overline{d-p}) = \log\left(\frac{1}{\kappa}\right) = -\log \kappa, \quad e^{\overline{d-p}} = \frac{1}{\kappa} - 1, \quad \overline{d-p} = \log\left(\frac{1}{\kappa} - 1\right).$$

Therefore,

$$\kappa_0 \equiv f(\overline{d-p}) - (1-\kappa)\overline{d-p} = -\log \kappa - (1-\kappa) \log\left(\frac{1}{\kappa} - 1\right).$$

We then have $\log(1+e^{d_{t+1}-p_{t+1}}) = f(d_{t+1} - p_{t+1}) \approx \kappa_0 + (1-\kappa)(d_{t+1} - p_{t+1})$, so

$$\begin{aligned} r_{e,t+1} &= p_{t+1} + \log\left(1+e^{d_{t+1}-p_{t+1}}\right) - p_t \\ &\approx p_{t+1} + \kappa_0 + (1-\kappa)(d_{t+1} - p_{t+1}) - p_t \\ &= \kappa_0 + (1-\kappa)d_{t+1} + \kappa p_{t+1} - p_t \\ &= \kappa_0 + \left[(1-\kappa)d_{t+1} - d_t + \kappa p_{t+1} - p_t + d_t\right] \\ &= \kappa_0 + (d_{t+1} - d_t) + \kappa(p_{t+1} - d_{t+1}) - (p_t - d_t) \\ &= \kappa_0 + \Delta d_{t+1} + \kappa(p_{t+1} - d_{t+1}) - (p_t - d_t). \end{aligned}$$

Therefore, we have

$$\begin{aligned} r_{e,t+1} &\approx \kappa_0 + \Delta d_{t+1} + \kappa(p_{t+1} - d_{t+1}) - (p_t - d_t), \\ \kappa &= \frac{1}{1+e^{\overline{d-p}}}, \\ \kappa_0 &= -\log(\kappa) - (1-\kappa) \log\left(\frac{1}{\kappa} - 1\right). \end{aligned}$$

Isolating $p_t - d_t$ and recursively substituting for $p_{t+h} - d_{t+h}$, we have

$$\begin{aligned} (p_t - d_t) &\approx \kappa_0 - r_{e,t+1} + \Delta d_{t+1} + \kappa(p_{t+1} - d_{t+1}) \\ &= \kappa_0 - r_{e,t+1} + \Delta d_{t+1} + \kappa \left[\kappa_0 - r_{e,t+2} + \Delta d_{t+2} + \kappa(p_{t+2} - d_{t+2}) \right] \\ &= (1+\kappa)\kappa_0 + (\Delta d_{t+1} + \kappa \Delta d_{t+2}) - (r_{e,t+1} + \kappa r_{e,t+2}) + \kappa^2(p_{t+2} - d_{t+2}) \\ &\vdots \text{ keep substituting } p_{t+h} - d_{t+h} \\ &= \frac{\kappa_0(1-\kappa^H)}{1-\kappa} + \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} - \sum_{h=1}^H \kappa^{h-1} r_{e,t+h} + \kappa^H(p_{t+H} - d_{t+H}). \end{aligned}$$

Thus, we have

$$d_t - p_t \approx \frac{\kappa_0(\kappa^H - 1)}{1-\kappa} + \sum_{h=1}^H \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} + \kappa^H(d_{t+H} - p_{t+H}),$$

so, letting $dp_t \equiv d_t - p_t$,

$$dp_t \approx \frac{\kappa_0(\kappa^H - 1)}{1 - \kappa} + \sum_{h=1}^H \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} + \kappa^H dp_{t+H}. \quad (1.1)$$

Taking expectations as of time t ,

$$dp_t = \frac{\kappa_0(\kappa^H - 1)}{1 - \kappa} + \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t[\Delta d_{t+h}] + \kappa^H \mathbb{E}_t[dp_{t+H}]. \quad (1.2)$$

Suppose $H \rightarrow \infty$. Since $\kappa = \frac{1}{1 + e^{\bar{d}-p}} \in (0, 1)$, we have $\lim_{H \rightarrow \infty} \kappa^H = 0$, and by the no-bubble condition $\lim_{H \rightarrow \infty} \kappa^H \mathbb{E}_t[dp_{t+H}] = 0$. Therefore,

$$dp_t = \frac{-\kappa_0}{1 - \kappa} + \sum_{h=1}^{\infty} \kappa^{h-1} \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^{\infty} \kappa^{h-1} \mathbb{E}_t[\Delta d_{t+h}]. \quad (1.3)$$

1.2 Part (b)

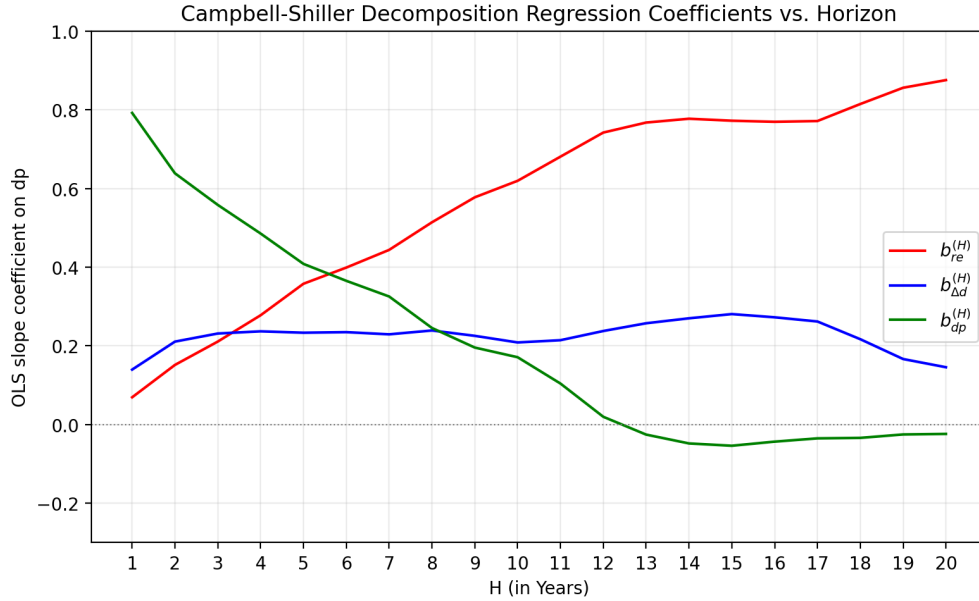


Figure 1: OLS slope coefficients of `future_return`, `change_d`, and `dp_H` on `dp`, plotted against the horizon H (in years).

From the equation

$$1 = \underbrace{\frac{\text{Cov} \left[dp_t, \sum_{h=1}^H \kappa^{h-1} r_{e,t+h} \right]}{\text{Var}[dp]}}_{b_{re}^{(H)}} + \underbrace{\frac{\text{Cov} \left[dp_t, -\sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} \right]}{\text{Var}[dp]}}_{b_{\Delta d}^{(H)}} + \underbrace{\frac{\text{Cov} [dp_t, \kappa^H dp_{t+H}]}{\text{Var}[dp]}}_{b_{dp}^{(H)}}.$$

This equation is trying to measure, how much each component moves when today's dp_t moves. The meaning of these three coefficients is as follows:

$$\sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t[r_{e,t+h}] = a + b_{r_e}^{(H)} dp_t + \epsilon_t$$

The slope is $b_{r_e}^{(H)}$, and it measures the sensitivity of expected discounted future returns to today's dividend-price ratio.

Because dividend growth enters the Campbell–Shiller equation with a minus sign, define

$$-\sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t[\Delta d_{t+h}] = a + b_{\Delta d}^{(H)} dp_t + \epsilon_t.$$

So $b_{\Delta d}^{(H)}$ captures how much variation in dp_t is associated with expected future dividend growth.

$$\kappa^H \mathbb{E}_t[dp_{t+H}] = a + b_{dp}^{(H)} dp_t + \epsilon_t.$$

$b_{dp}^{(H)}$ tells us how much today's high dp_t is simply expected to persist into the future.

$$1 = b_{r_e}^{(H)} + b_{\Delta d}^{(H)} + b_{dp}^{(H)}$$

From the graph, we can see that the contributions of expected future returns, expected future dividend growth, and the future dividend-price ratio to the variation in today's dp_t change with the horizon H . In the short run, the variation in today's dp_t is primarily driven by the future dividend-price ratio. As the horizon H increases, the contribution of expected future returns becomes more significant, while the contribution of the future dividend-price ratio declines toward zero. The contribution of negative expected future dividend growth remains relatively stable across different horizons. Also, if we add up these three coefficients, we get approximately 1, which reflects the normalized variance decomposition of today's dp_t .

1.3 Part (c)

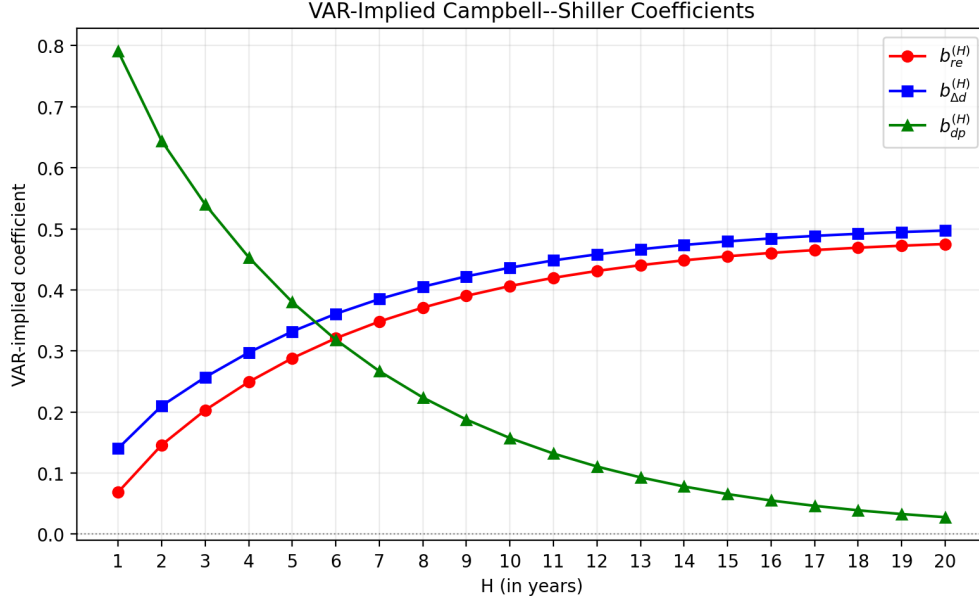


Figure 2: VAR-implied coefficients

The VAR-implied coefficients shown in Figure 2 shows some interesting patterns. Instead of only the component of expected future returns dominates as H increases, it is shown that both the component of expected future returns and the component of negative expected future dividend growth contribute significantly to the variation in today's dp_t across different horizons. At the same time, the component of the future dividend-price ratio declines as the horizon increases, consistent with, and seems to be greater than the findings from the direct-regression estimates. Note that if we add up these three coefficients, we still get approximately 1, which reflects the normalized variance decomposition of today's dp_t . The difference in the patterns may arise because the two approaches estimate long-horizon coefficients differently. The direct approach estimates each horizon separately using realized future outcomes, whereas the VAR approach estimates the joint one-period dynamics of the state variables and uses the estimated transition matrix to imply coefficients at all subsequent horizons.

1.4 Part (d)

The resulting long-run VAR-implied coefficients are reported in Table 1. From the table, we can see that in the long run, the contributions of expected future returns and negative expected future dividend growth are roughly equal, and their sum is approximately 1, which is consistent with the normalized variance decomposition of today's dp_t .

Table 1: Long-run VAR-implied coefficients

Component	Coefficient	Estimate
Expected return	$b_{re}^{(\infty)}$	0.489191
Negative expected dividend growth	$b_{dg}^{(\infty)}$	0.509617
Sum	$b_{re}^{(\infty)} + b_{dg}^{(\infty)}$	0.998808

2 Question 2

3 Question 3

4 Question 4